UAM | ESCUELA POLITÉCNICA SUPERIOR

Programa Oficial de Posgrado

Propuesta de proyecto del Trabajo Fin de Máster (TFM) /Proposal of Project for the Master's Thesis [1]

Máster Universitario/Master's degree: Máster Universitario en Sistemas Interactivos Inteligentes

Título/*Title*: Predicción de la expansión de variantes de SARS-CoV-2 en la población mediante Aprendizaje Automático / Predicting the expansion of SARS-CoV-2 variants in the population using Machine Learning

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Curso/*Course year: 2025-26*

Fecha/*Date:* 10-06-2026

The different secDons of this document must be wriHen either in Spanish or in English, depending on the language in which the Master's degree is taught. The document should have an indicaDve length of between two and four pages. It must be sent to gesDonmaster.eps@uam.es within the deadlines set for this purpose in each academic year.

1.Introducción/*Introduction*

The COVID-19 pandemic exposed a structural limitation of conventional genomic surveillance: systems based on consensus sequences detect the emergence of new variants only once they already circulate dominantly in the population, leaving an insufficient response window to adapt vaccines and treatments. SARS-CoV-2 is, however, the most extensively sequenced virus in history, with more than 17 million genomes deposited in GISAID together with metadata on collection date, geographic location and phylogenetic classification, which makes it an ideal setting to explore the computational prediction of its spread.

Recent work has approached this problem from different angles. PandoGen (Ramachandran et al., 2024) showed that autoregressive generative models trained on Spike sequences can anticipate variants months in advance. VirusT5 (Marathe et al., 2026) reframed viral evolution as a sequence-to-sequence translation problem, reaching predicted identities above 99%. Despite their promising results, these approaches operate exclusively on population-level consensus sequences and do not incorporate intra-host diversity information.

This work builds on a previous biological observation by the group: within the intra-host mutant spectra of COVID-19 patients there are minority haplotypes carrying mutations characteristic of clades that have not yet emerged in the population at the time of sampling, the so-called clade-discordant haplotypes (de Ávila et al., 2025; Martínez-González et al., 2022). Their existence suggests that intra-host viral diversity may contain anticipatory signals of the virus's future spread in the population.

The aim of this thesis is to develop and validate a machine learning model able to predict the spread of SARS-CoV-2 clades in the population from global genomic surveillance metadata, quantifying the anticipation time through independent biological validation with haplotypes identified from ultra-deep sequencing data of 28 patients distributed across six pandemic waves. The work integrates bioinformatics, viral evolution and machine learning applied to real data, which places it naturally within the scope of the Master?s in Interactive Intelligent Systems

*2.*Objetivos y planteamiento /*Objectives and approach*

The general aim of this Master's thesis is to build a predictive model able to anticipate the spread of SARSCoV-2 clades in the population from global genomic surveillance data, and to quantify its anticipation capacity through independent biological validation.

The following specific objectives are defined, ordered according to the logic of the developed pipeline:

O1. Construction of the training dataset. Process and filter the metadata of approximately 17 million GISAID sequences, encoding genomic and temporal information per clade and time window as predictive features.

O2. Development of the predictive model. Train a machine learning classifier with walk-forward validation to predict which clades will spread and become dominant, respecting the temporal structure of the data and avoiding information leakage between waves.

O3. Evaluation of predictive performance. Measure the model's discriminative ability through AUC and its temporal consistency across multiple validation windows.

O4. Interpretability. Apply Explainable AI (XAI) techniques to interpret the model in an intuitive way and understand which features are most informative for the prediction.

3. Metodología y plan de trabajo / Methodology and work plan

The Master's thesis follows an applied-research orientation with a software-development component, implemented entirely in Python. The pipeline is organized into sequential modules covering everything from data ingestion to the final biological validation.

The main data source is the GISAID (Global Initiative on Sharing All Influenza Data) metadata file in TSV format. GISAID is the largest public database of viral genomes in the world and contains approximately 17 million SARS-CoV-2 sequences annotated with collection date, geographic location, Nextstrain clade, Pango lineage and amino acid substitutions. Given the data volume, processing is carried out by reading in chunks to avoid memory limitations in the execution environment.

The predictive model is based on a machine learning classifier optimized for high-dimensional, sparse tabular data. Validation follows a strict walk-forward scheme in which the model is trained exclusively on data prior to each prediction window, faithfully reproducing real-use conditions and avoiding any information leakage between pandemic waves.

Tareas y subtareas / *Tasks and subtasks*	**Horas / *Hours***
T1. Literature review	**28**
T1.1 Viral evolution and genomic surveillance	14
T1.2 Machine learning applied to viruses	14
T2. GISAID data acquisition and preprocessing	**48**
T2.1 Metadata download and filtering	15
T2.2 Substitution parsing and normalization	18
T2.3 Construction of temporal windows per clade	15
T3. Feature engineering	**43**
T3.1 Per-position encoding of substitutions	20
T3.2 Construction of the target variable	23
T4. Predictive model development	**53**
T4.1 Training with walk-forward validation	28
T4.2 Tuning and feature selection	25
T5. Model interpretability	**25**
T5.1 Feature importance analysis per clade	12
T5.2 Visualization and interpretation of results	13
T6. Biological validation with UDS data	**45**
T6.1 Integration of haplotypes from 28 patients (6 waves)	20
T6.2 Lead-time computation and statistical analysis	25
T7. Manuscript writing	**48**
T7.1 Writing, figures and tables	30
T7.2 Review with advisor and corrections	18
T8. Defense preparation	**10**
T8.1 Slides preparation	5
T8.2 Rehearsal and defense	5
TOTAL DE HORAS / *TOTAL HOURS*	**300**

4.BibliograDía inicial / Initial bibliography

1. de Ávila AI, Soria ME, Martínez-González B, Somovilla P, Mínguez P, Salar-Vidal L, et al. SARS-CoV-2 biological clones are genetically heterogeneous and include clade-discordant residues. J Virol. 2025;99(5): e0225024.

2. Martínez-González B, Soria ME, Vázquez-Sirvent L, Ferrer-Orta C, Lobo-Vega R, Mínguez P, et al. SARS-CoV-2 Mutant Spectra at Different Depth Levels Reveal an Overwhelming Abundance of Low Frequency Mutations. Pathogens. 2022;11(6):662.

3. Martínez-González B, Soria ME, Vázquez-Sirvent L, Ferrer-Orta C, Lobo-Vega R, Mínguez P, et al. SARS-CoV-2 Point Mutation and Deletion Spectra and Their Association with Different Disease Outcomes. Microbiol Spectr. 2022;10(2):e0022122.

4. Delgado S, Somovilla P, Ferrer-Orta C, Martínez-González B, Vázquez-Monteagudo S, Muñoz-Flores J, et al. Incipient functional SARS-CoV-2 diversification identified through neural network haplotype maps. Proc Natl Acad Sci USA. 2024;121(10):e2317851121.

5. Ferrer-Orta C, Vázquez-Monteagudo S, Ferrero DS, Martínez-González B, Perales C, Domingo E, et al. Point mutations at specific sites of the nsp12-nsp8 interface dramatically affect the RNA polymerization activity of SARS-CoV-2. Proc Natl Acad Sci USA. 2024;121(29):e2317977121.

6. Ramachandran P, Vasan K, Lensink MF, Thorne LH. PandoGen: predicting future COVID-19 sequences. PLOS Comput Biol. 2024;20(1):e1011790.

7. Marathe V, et al. VirusT5: predicting SARS-CoV-2 spike evolution using a transformer-based translation model. IEEE. 2026.

8. Thadani NN, Gurev S, Notin P, Youssef N, Rollins NJ, Ritter A, et al. Learning from prepandemic data to forecast viral escape. Nature. 2023;622:818–825.

Han W, Chen N, Xu X, et al. Predicting the antigenic evolution of SARS-CoV-2 with deep learning. Nat. Commun. 2023;14:3478.

9.Dadonaite B, Brown J, McMahon TE, et al. Spike deep mutational scanning helps predict success of SARS-CoV-2 clades. Nature. 2024;631:617–626.

5. Propuesta de tribunal / Evaluation panel proposal

El tribunal o comisión evaluadora estará formado por tres miembros, ninguno de ellos podrá ser director o ponente del TFM.
Se solicita seis posibles miembros para formar el tribunal. Todos los miembros propuestos deberán tener una Btulación igual o superior a la del máster y haber dado su conformidad para formar parte, si procede, del tribunal definiBvo. Al menos tres de los miembros propuestos serán externos al grupo de invesBgación del director/es y ponente. En el MUBBC no es necesario contactar ni solicitar su conformidad a los miembros propuestos del tribunal en el momento de enviar el proyecto.
En cualquier caso, será la comisión de coordinación y seguimiento del máster quien defina la composición del tribunal (Btulares y suplentes), pudiéndose dar la situación de que algún miembro del tribunal definiBvo no esté entre los propuestos en esta solicitud.
The evaluaDon panel will be made up of three members, none of whom may be the advisor nor the lecturer of the Master's thesis.
Six possible members are requested to form the evaluaDon panel. All the proposed members must have a degree equal to or higher than that of the master's degree and must have given their agreement to form part, if applicable, of the final evaluaDon panel. At least three of the proposed members must be external to the research group of the advisor(s) and lecturer. In the MUBBC it is neither necessary to contact nor request the agreement of the proposed members of the evaluaDon panel at the Dme of submiVng the project.

In any case, the Master's CoordinaDon and Monitoring CommiHee will be responsible for defining the composiDon of the panel (members and subsDtutes), and it is possible that some members of the final panel may not be among those proposed in this applicaDon form.

Propuesta 1 / First proposed member

Nombre y Apellidos / Name and surname: Pablo Varona Martínez

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Propuesta 6 / Sixth proposed member

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El estudiante/*The student*

Fdo./*Signed*: SERGIO LÓPEZ GONZÁLEZ DEL REY

<u>Vº Bº del director o directores/*Approval of the advisor(s)*</u>

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